

Mathematics

A Cultural Heritage

Lecture 8 (part 2)

Proofs

Course Topics

*Funny Side of
Mathematics*

Mathematics,
literatures, movies,
and more

**Mathematics
and the Arts**

**Mathematics
and Education**

Two mathematics
cultures and a few
mathematical lives

Numbers

Numbers
at work

Our Obsessions
with Numbers

**Mathematics
and Society**

Mathematics and Truth
(Philosophical Aspects
of Mathematics)

**THEOREMS (PROBLEMS)
AND PROOFS**

Truth and Observations

1. Patterns were observed (like Pythagorean triplets).
2. Throughout history, many observations led to “guesses”.
3. In mathematics, we use “proof” to remove doubt (up to some basic assumptions and basic logic).

Truth and proofs

1. Without proof, truth is just an opinion.
2. You are trusting an authority.
3. Why should you trust this authority?

General Perception of Math as Truth

Video: $2 + 2 = \dots$



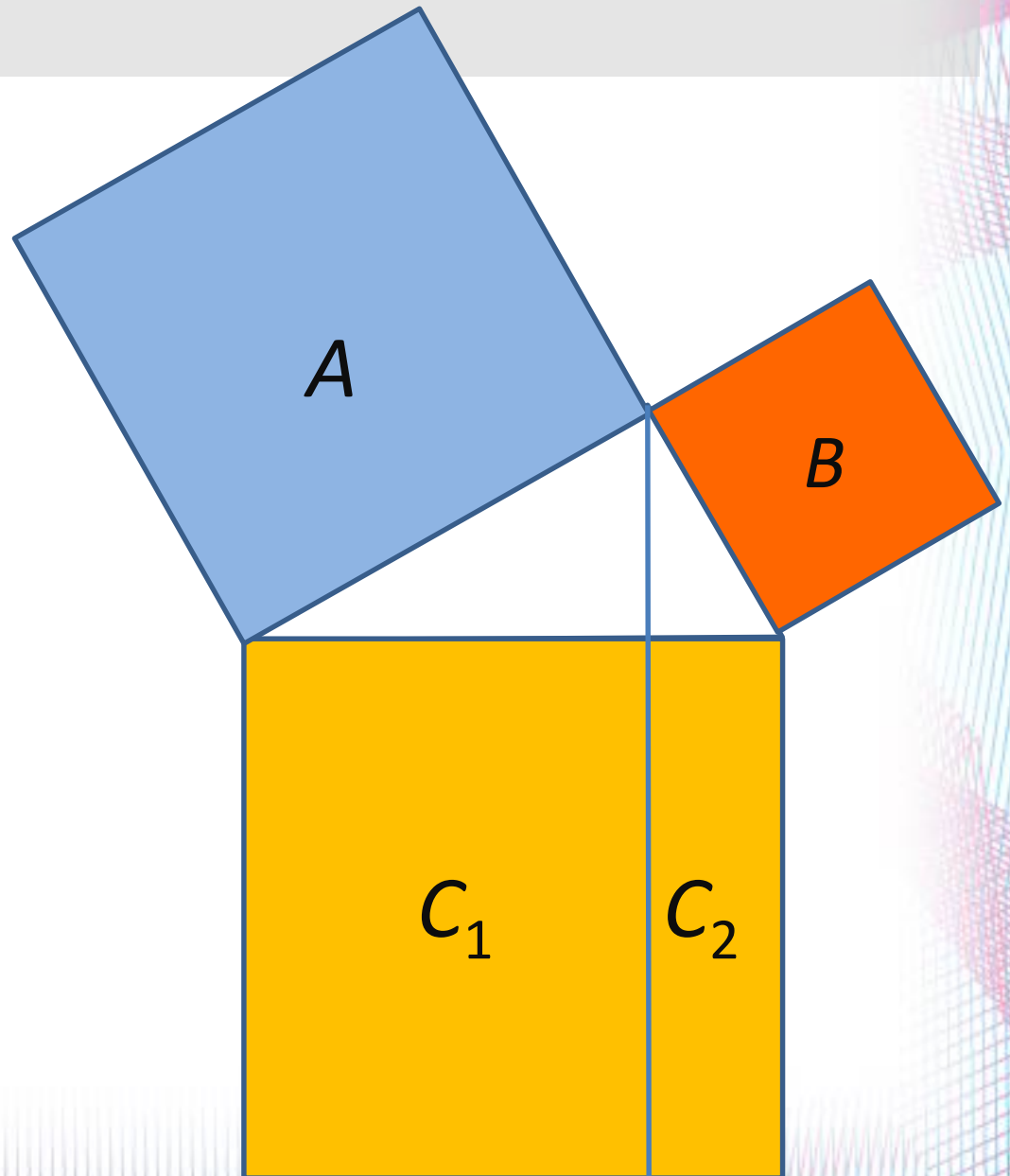
The Proof in Euclid's *Elements*

Strategy of proof

- Decompose area C into sub-areas C_1 and C_2
- Show that areas $A = C_1$, $B = C_2$

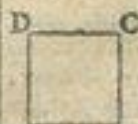
Hence:

$$A + B = C_1 + C_2 = C$$



PROBL. 14. PROPOS. 46.

A DATA recta linea quadratum describere.



Si data recta AB, sup qua oporteat quadratum describere. Ex A, & B, educantur AD, BC, perpendiculares ad AB, sicut ipsi A, B, æquales, & connectatur recta CD. Dico AB, CD, esse quadratum. Cum, n. anguli A, & B, sint recti, erunt AD, BC, parallele: Sunt aut & æquales, quod utraq; æqualis sit ipsi AB; Igitur & AB, DC, parallele sunt & æquales: & ideo parallelogrammum est ABCD; in quo, cum AD, DC, CB, æquales sint ipsi AB, omnes quatuor linee æquales existunt; Sunt aut & oēs quatuor anguli recti, cum C, & D, æquales sint oppositis rectis A, & B. Quadratum igitur est ABCD, ex definitione. Ac pende a data recta linea quadratum describitur; Quod faciendum erat.

EX PROCLO.

LINEARVM æqualium æqualia sunt quadrata: & quadratorum æqualium æquales sunt linee.



Si n. primo recte AB, CD, æquales; Dico earum quadrata ABFG, EDCB, æqualia quoque esse. Ductis enim diametris BF, DE, erunt duo latera BA, AF, trianguli BAF, duobus lateribus DC, CE, trianguli DCE, æqualia, verumq; recti; & ex definitione quadrati recte AF, CE, æquales sint rectis AB, CD: Sunt autem, & anguli A, & C, æquales, nempe recti. Igitur triangula BAF, DCE, æqualia erunt. Quæ cum sint dimidia quadratorum, erunt & quadrata tota æqualia. Quod est propositum.



Sunt secundo quadrata ABDE, BCFG, æqualia; Dico lineas quoque ipsorum AB, BC, æquales esse. Cōiungantur n. quadrata ad angulum B, ut recte AB, BC, in directum cōiungantur. Et quoniam anguli ABG, CBD, sunt recti, erunt & recte GB, BD, in directum cōiunctæ. Ducatur diametri AD, CF, iunganturq; recte AG, CD. Quoniam igitur quadrata ABDE, BCFG, æqualia sunt, erunt & triangula ABD, BCG, eorū dimidia, æqualia. Addito ergo eōi triangulo BCD, fiet totū triangulum ACD, æqualia. Addito ergo eōi triangulo BCD, fiet totū triangulum ACD, æqualia. Quare triangula ACD, GDC, eōi eandem

dem habent basin CD, ad easdemq; sint partes, in easdem sunt parallelis, ideoq; parallele sunt AG, CD. Et qm, ut in scholio propof. 14, ostēditur, diameter in quadrato secat angulos quadrati bisariam, erunt anguli DAC, GCA, alteri semirecti, ideoq; æquales. Quoniam & parallele sunt AD, CG, igitur parallelogrammum est ADCG, ac propterea recte AD, CG, æquales: Quoniam ergo in triangulis ABD, BCG, latera AD, CG, æqualia sunt, & anguli, qd ea latera adiacenti, inter se & æquales, cum sint semirecti, ut in scholio propof. 14, ostēditur fuit, erunt itaque latera equalia, nempe AB, ipsi BC, & qd e ppositū.

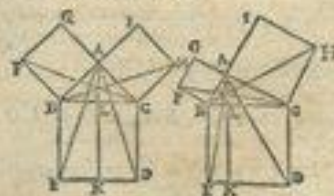
SCHOLION.

POSSUNT hæc omnia multo brevius probari per superpositum quadrati unius super aliud. Nā si linea sunt æquales, si una alteri superponatur, congruent ipsa inter se. Cum ergo & anguli sint æquales, nempe recti, conveniēt quoq; ipsi inter se, ideoq; totū quadratum toti quadrato congruit. Quod si quadrata sunt æqualia, congruent ipsa inter se, propter æqualitatem angulorum, & qm & triangula totū quadratum aliud mutant ipsa.

THEOR. 33. PROPOS. 47.

IN rectangulis triangulis, quadratum, quod a latere rectum angulum subtendente describitur, æquale est eis, quæ a lateribus rectum angulum continentibus.

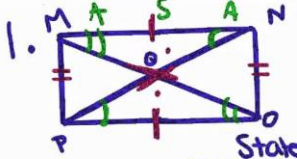
In triangulo ABC, angulus BAC, sit rectus, describaturq; super AB, AC, BC, quadrata ABFG, ACHI, BCDE. Dico quadratum BCDE, descriptum super latere BC, quod angulo recto opponit, æquale esse duobus quadratis ABFG, ACHI, quæ super alia duo latera sunt descripta, siue hæc duo latera æqualia sint, siue inæqualia.



Ducatur enim recta AK, parallela ipsi BE, vel ipsi CD, secans BC, in L, & iungantur recte AD, AE, CF, BH. Et quia duo anguli BAC, & BAG, sunt recti, erunt recte GA, AC, una linea recta; eodēq; modo I A, AB, una linea erunt.

Truth and Proofs

1. A proof is a logical series of steps that goes from an assumption A to a conclusion B.
2. Each step has a clear reason/explanation.
3. No gaps, no additional assumptions!



Given: $MNOP$ is a rectangle
Prove: $\triangle MQN \cong \triangle POQ$

Statement	Reason
1. $MNOP$ is a $\square$	1. Given
2. $\overline{MN} \cong \overline{PO}$	2. If $\square$, then opp sides $\cong$
3. $\overline{MN} \parallel \overline{PO}$ $\overline{MP} \parallel \overline{NO}$	3. If $\square$, then opp sides $\parallel$
4. $\angle MQN \cong \angle POQ$ $\angle NMQ \cong \angle OPQ$	4. If lines $\parallel$, then alt. int. $\angle$'s $\cong$
5. $\triangle MQN \cong \triangle POQ$	5.

Theorem 29 (Commutative Law of Multiplication):

$$xy = yx.$$

Proof: Fix y , and let $\mathfrak{M}$ be the set of all x for which the assertion holds.

I) We have

$$y \cdot 1 = y,$$

and furthermore, by the construction in the proof of Theorem 28,

$$1 \cdot y = y,$$

hence

$$1 \cdot y = y \cdot 1,$$

so that 1 belongs to $\mathfrak{M}$.

II) If x belongs to $\mathfrak{M}$, then

$$xy = yx,$$

hence

$$xy + y = yx + y = yx'.$$

By the construction in the proof of Theorem 28, we have

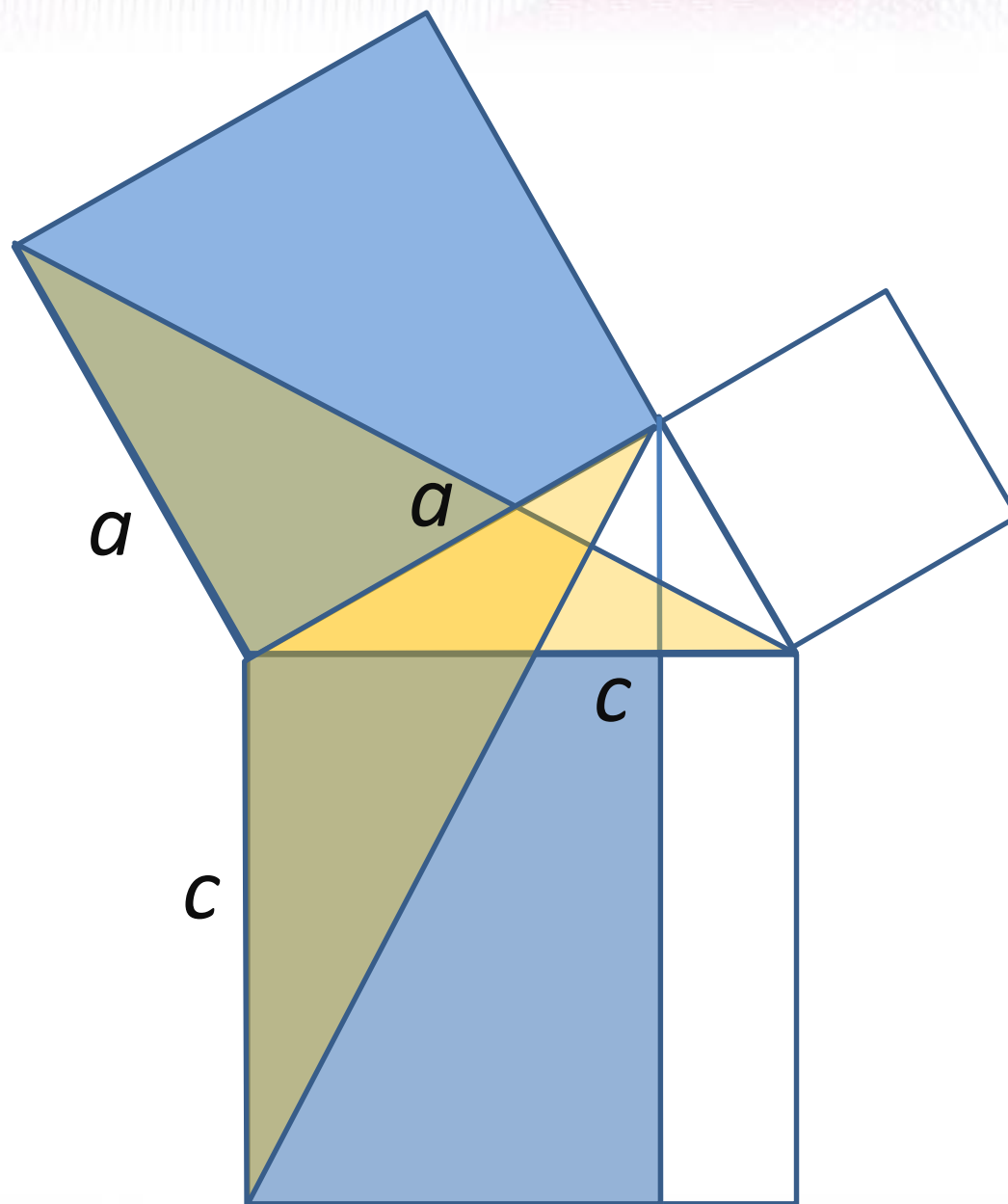
$$x'y = xy + y,$$

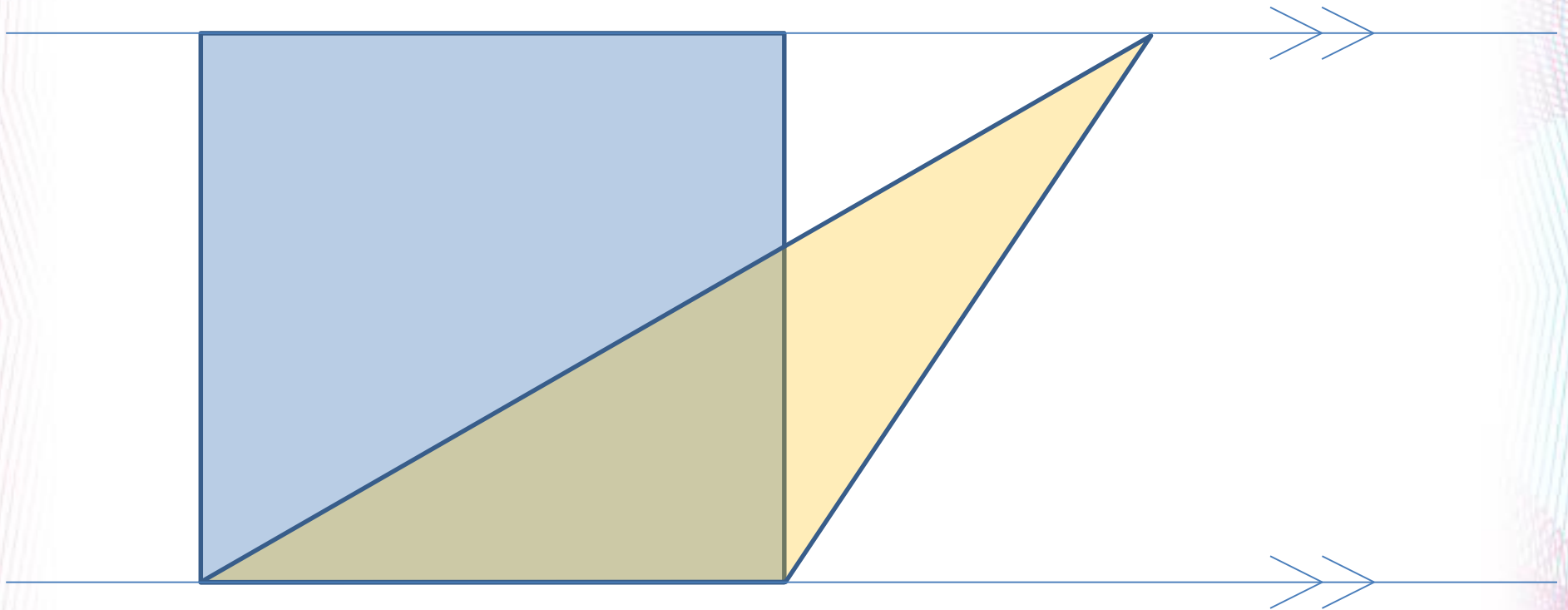
hence

$$x'y = yx',$$

so that x' belongs to $\mathfrak{M}$.

The assertion therefore holds for all x .

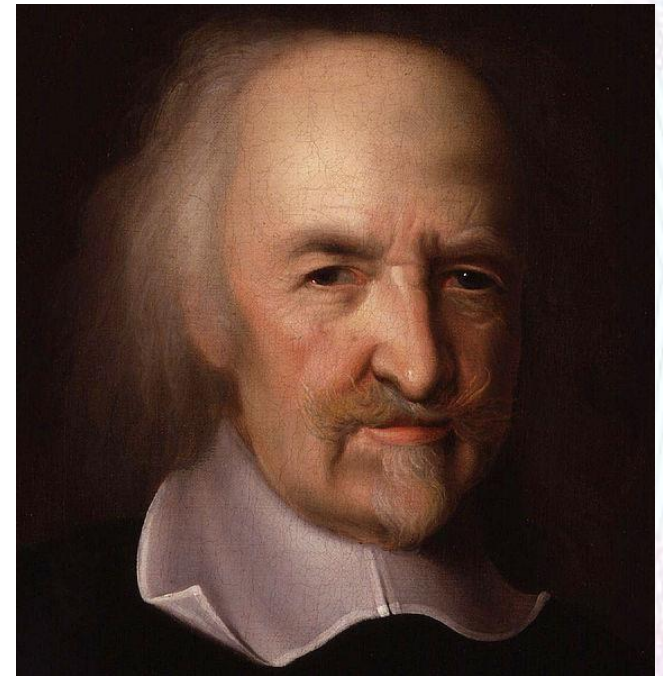




Area of rectangle = 2 x Area of triangle

How Thomas Hobbes Learned Geometry

1. **Thomas Hobbes** learned Euclid's *Elements* at the age of forty, only after looking at Proposition 47 in Book I of *Elements*.

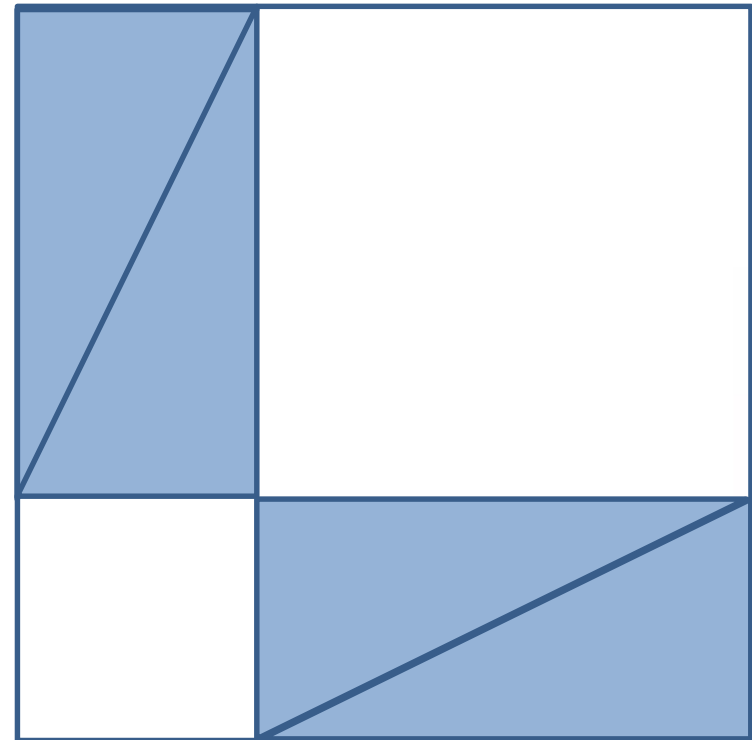
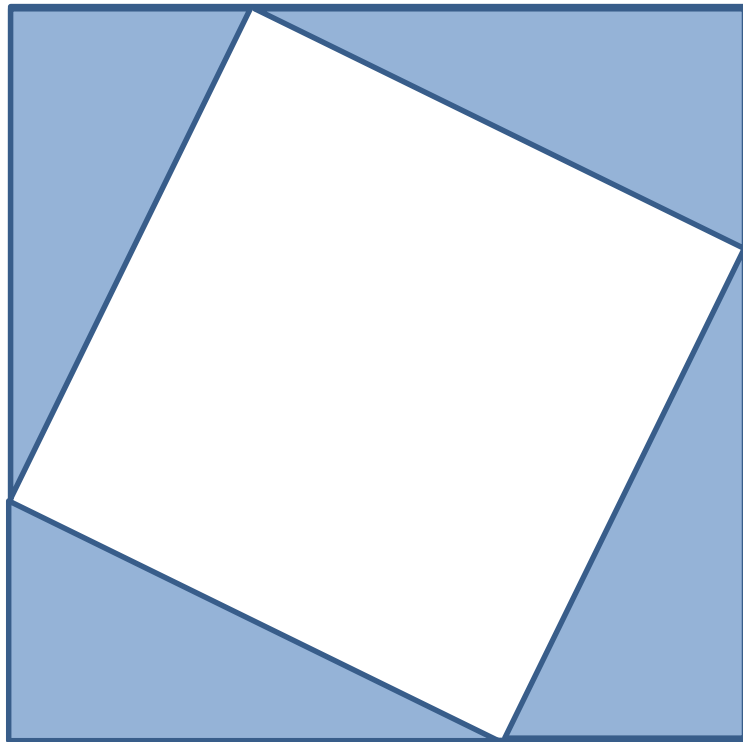


Thomas Hobbes (1588-1679)

Source: wikipedia

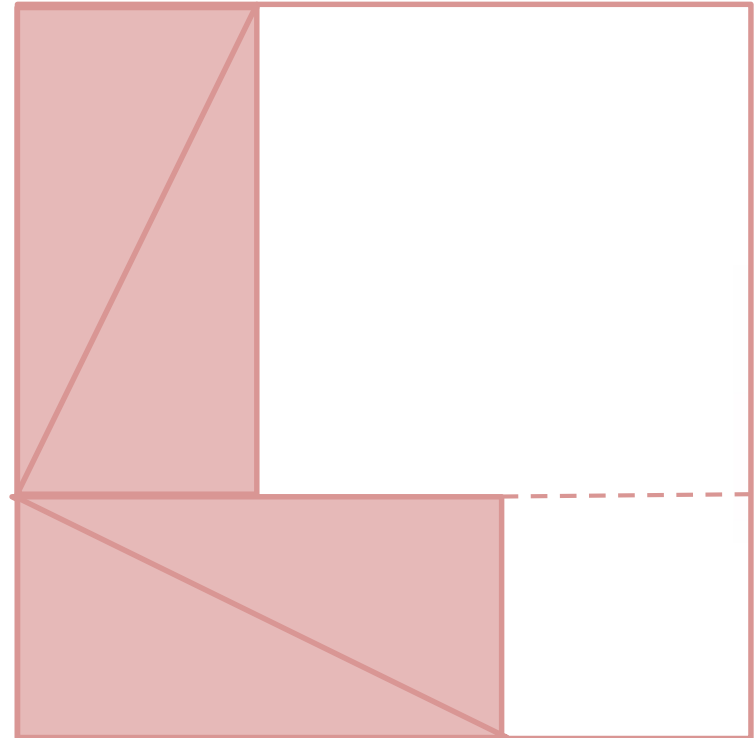
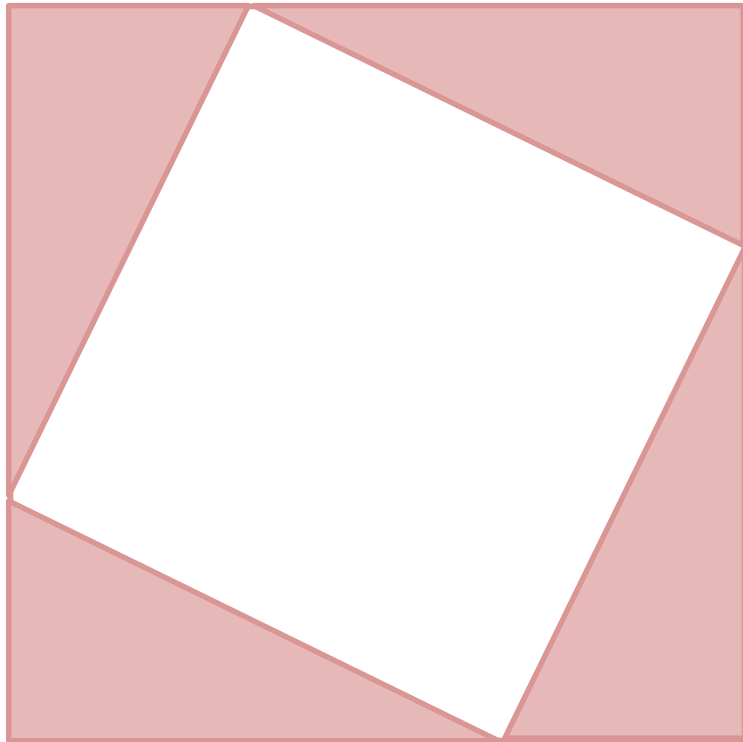
A Chinese Proof

1. The following proof was given by 趙爽(Zhao Shuang) in the 3rd Century.



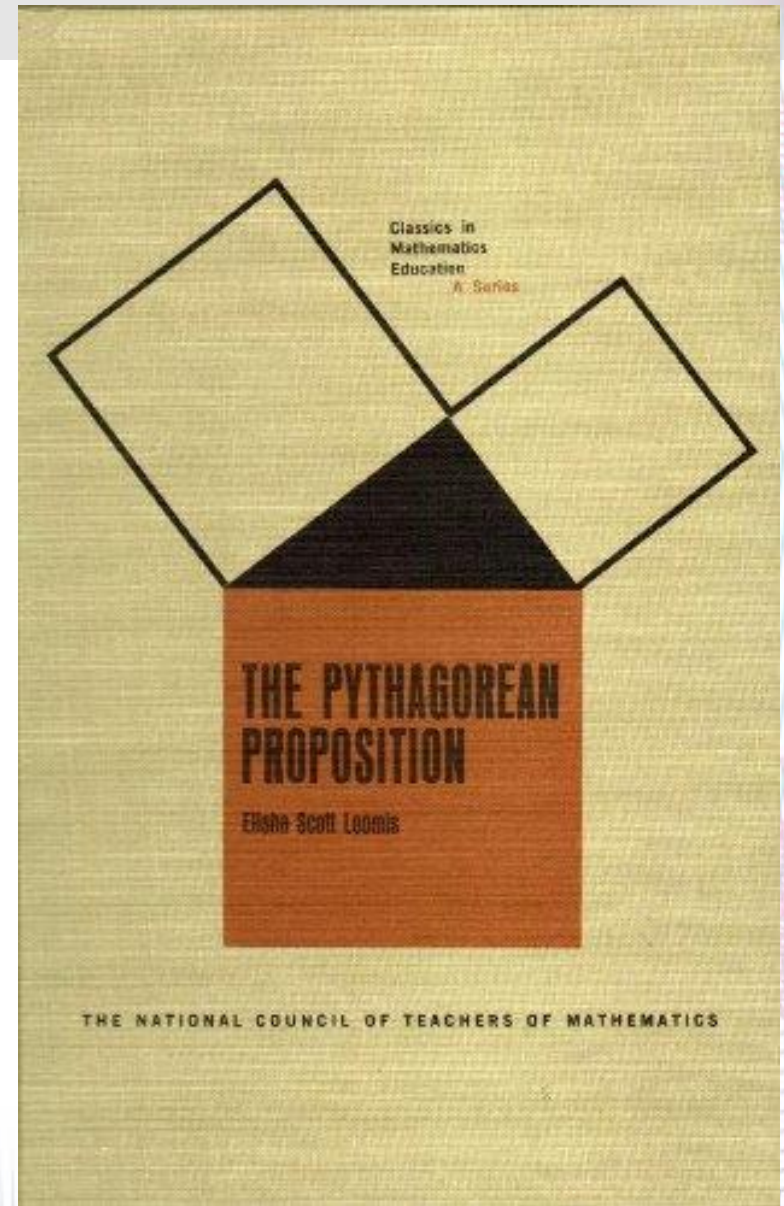
Another Proof

1. Here is another proof using a similar approach.

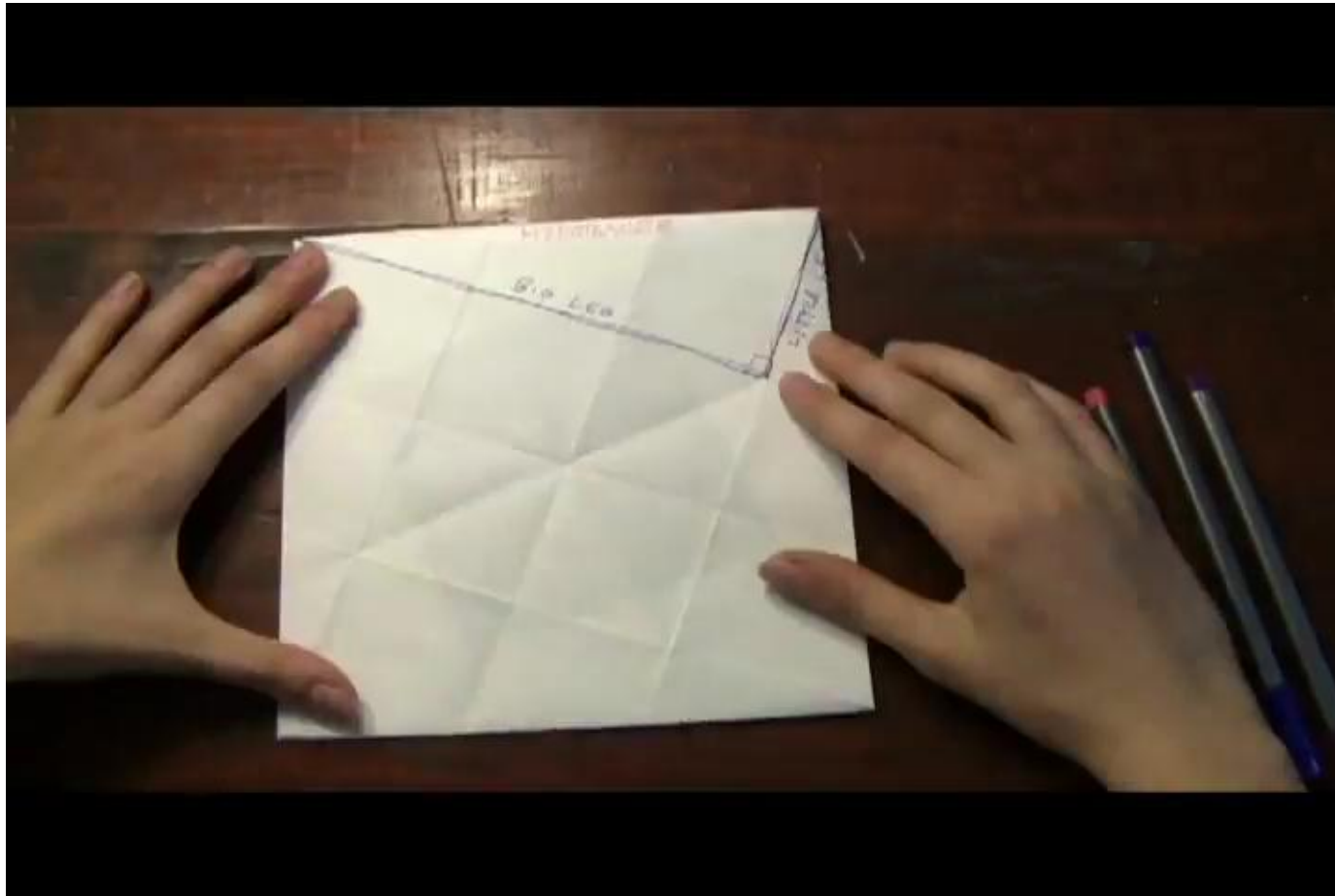


Many Proofs of the Theorem

1. The book *The Pythagorean Proposition* compiled by Elisha Scott Loomis in 1927 collected 370 different proofs of the Pythagoras Theorem.



Video: Origami Proof of the Pythagorean Theorem

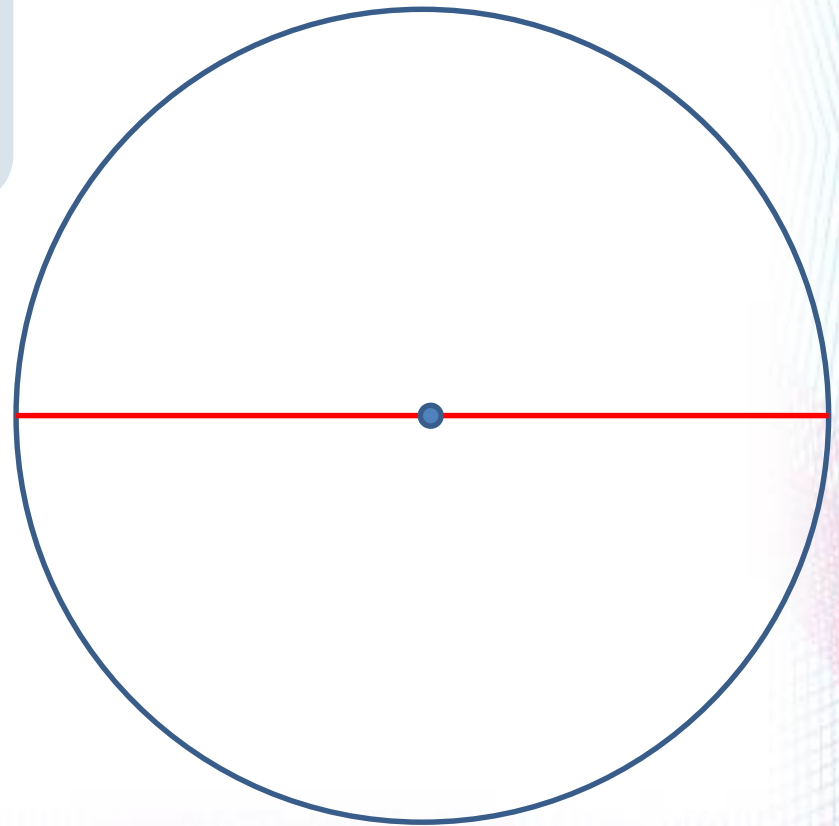


Source: www.youtube.com/watch?v=z6lL83wl31E

The First Ever Mathematical Proof

1. **Thales** of Miletus (~600 BC) proved that:

Theorem The diameter divides a circle into two equal parts.



Why Do We Need Proofs ???

Why Do We Need Proofs

Is it to certify that a statement is true?

To convince others (and ourselves) that a statement is true?

A ritual practiced by a cult, the members of which are called mathematicians?

To understand, and to gain deeper understanding?